

Conditional statements and loops questions

1. What will be the output of the following code?

```
s = "madam"
```

```
if s == s[::-1]:  
    if len(s) % 2 == 0:  
        print("Even Length Palindrome")  
    else:  
        print("Odd Length Palindrome")  
else:  
    print("Not Palindrome")
```

- A) Even Length Palindrome
- B) Odd Length Palindrome
- C) Not Palindrome
- D) Error

Correct Answer: B) Odd Length Palindrome

2. What will be the output of the following code?

```
for i in range(1, 6):  
    if i % 3 == 0:  
        print("Fizz", end=" ")  
    elif i % 5 == 0:  
        print("Buzz", end=" ")  
    else:  
        print(i, end=" ")
```

- A) 1 2 Fizz 4 Buzz
- B) 1 2 Fizz 4 5
- C) 1 2 3 4 Buzz
- D) Fizz Buzz 1 2 3

Correct Answer: A) 1 2 Fizz 4 Buzz

3.What will be the output of the following code?

```
n = 123
sum = 0
while n > 0:
    sum += n % 10
    n = n // 10
print(sum)
```

Options:

- A) 6
- B) 123
- C) 5
- D) 9

Correct Answer: A) 6

4.What will be the output of the following code?

```
n = 5
for i in range(2, n):
    if n % i == 0:
        print("Not Prime")
        break
else:
    print("Prime")
```

- A) Prime
- B) Not Prime
- C) Error
- D) No Output

Correct Answer: A) Prime

Data type Questions + comparison operators

5.What will be the output of the following code?

```
x = 10
```

```
y = "5"
```

```
z = x + int(y) + float(y)
```

```
print(type(z), z)
```

A) <class 'int'> 20

B) <class 'float'> 20.0

C) <class 'float'> 20

D) Error

Correct Answer: B) <class 'float'> 20.0

6.What will be the output of the following code?

```
a = 3
```

```
b = 3.0
```

```
c = a == b
```

```
d = type(a) == type(b)
```

```
print(c, d)
```

A) True True

B) False False

C) True False

D) False True

Correct Answer: C) True False

Inbuilt Data Structures Questions

7.What will be the output of the following code?

```
lst = ["Data", "Science", "AI"]
```

```
result = "-".join(lst)
```

```
print(result)
```

- A) Data Science AI
- B) Data-Science-AI
- C) ['Data-Science-AI']
- D) Error

Correct Answer: B) Data-Science-AI

8.What will be the output of the following code?

```
s = "apple,banana,mango"
```

```
lst = s.split(",")
```

```
print(len(lst), lst[1])
```

- A) 2 banana
- B) 3 banana
- C) 3 mango
- D) Error

Correct Answer: B) 3 banana

9.What will be the output of the following code?

```
lst = [1, 2, 3]
```

```
print(lst * 2)
```

- A) [2, 4, 6]
- B) [1, 2, 3, 1, 2, 3]
- C) [1, 4, 9]
- D) Error

Correct Answer: B) [1, 2, 3, 1, 2, 3]

10.What will be the output of the following code?

```
t = (1, 2, 3)
```

```
t[0] = 10
```

```
print(t)
```

- A) (10, 2, 3)
- B) (1, 2, 3)
- C) Error
- D) None

Correct Answer: C) Error

11. What will be the output of the following code?

```
s = {1, 2, 2, 3, 4}
```

```
print(len(s))
```

- A) 5
- B) 4
- C) 3
- D) Error

Correct Answer: B) 4

12. What will be the output of the following code?

```
d = {"a": 1, "b": 2}
```

```
print(d.get("c", 0))
```

- A) KeyError
- B) None
- C) 0
- D) Error

Correct Answer: C) 0

13. What will be the output of the following code?

```
x = {1, 2, 3}
```

```
y = {3, 4, 5}
```

```
print(x & y, x | y)
```

- A) {1, 2, 3, 4, 5} {3}
- B) {3} {1, 2, 3, 4, 5}

- C) {3} {3, 4, 5}
- D) Error

Correct Answer: B) {3} {1, 2, 3, 4, 5}

NUMPY Questions

14. What will be the output of the following code?

```
import numpy as np
```

```
arr = np.array([1, 2, 3])
```

```
print(arr * 2)
```

Options:

- A) [2, 4, 6]
- B) [1, 2, 3, 1, 2, 3]
- C) [1, 4, 9]
- D) Error

Correct Answer: A) [2, 4, 6]

15. What will be the output of the following code?

```
import numpy as np
```

```
arr = np.arange(2, 10, 3)
```

```
print(arr)
```

- A) [2 5 8]
- B) [2 4 6 8]
- C) [3 6 9]
- D) Error

Correct Answer: A) [2 5 8]

16.What will be the output of the following code?

```
import numpy as np
```

```
arr = np.ones((2, 2))
```

```
print(np.mean(arr))
```

- A) 0
- B) 1
- C) 2
- D) Error

Correct Answer: B) 1

17.What will be the output of the following code?

```
import numpy as np
```

```
arr = np.array([1, 2, 3, 4])
```

```
result = np.where(arr > 2, 10, 0)
```

```
print(result)
```

- A) [0 0 10 10]
- B) [1 2 10 10]
- C) [0 0 3 4]
- D) Error

Correct Answer: A) [0 0 10 10]

18.What will be the output of the following code?

```
import numpy as np
```

```
arr = np.array([1, 2, 3, 4])
```

```
new_arr = arr.reshape(2, 2)
```

```
print(new_arr)
```

Options:

- A) `[[1 2]`
 `[3 4]]`
- B) `[1 2 3 4]`
- C) `[[1 2 3 4]]`
- D) Error

Correct Answer: A) `[[1 2]`
 `[3 4]]`

PANDAS QUESTIONS

19.What will be the output of the following code?

```
import pandas as pd
```

```
s = pd.Series([10, 20, 30], index=["a", "b", "c"])
```

```
print(s["b"])
```

- A) 10
- B) 20
- C) 30
- D) Error

Correct Answer: B) 20

20.What will be the output of the following code?

```
import pandas as pd
```

```
df = pd.DataFrame({
```

```
    "A": [1, 2],
```

```
    "B": [3, 4]
```



```
})
```

```
print(df.shape)
```

Options:

- A) (2, 2)
- B) (1, 2)
- C) (2, 1)
- D) Error

Correct Answer: A) (2, 2)

21. What will be the output of the following code?

```
import pandas as pd
```

```
df = pd.DataFrame({  
    "A": [10, 20, 30],  
    "B": [40, 50, 60]  
})
```

```
print(df.iloc[1])
```

- A) A=10, B=40
- B) A=20, B=50
- C) A=30, B=60
- D) Error

Correct Answer: B) A=20, B=50

22. What will be the output of the following code?

```
import pandas as pd
```

```
df = pd.DataFrame({  
    "A": [1, 2, 3],  
    "B": [4, 5, 6]  
})
```

```
print(df.loc[0, "B"])
```

Options:

- A) 1
- B) 4
- C) 5
- D) Error

Correct Answer: B) 4

23.What will be the output of the following code?

```
import pandas as pd
```

```
df = pd.DataFrame({  
    "A": [1, 1, 2, 2],  
    "B": [10, 20, 30, 40]  
})
```

```
print(df.groupby("A")["B"].sum())
```

Options:

- A) 1 → 30, 2 → 70
- B) 1 → 10, 2 → 30
- C) 1 → 20, 2 → 40
- D) Error

Correct Answer: A) 1 → 30, 2 → 70

24. Which function is used to handle missing values in Pandas?

Options:

- A) `dropna()`
- B) `fillna()`
- C) Both A and B
- D) `remove()`

Correct Answer: C) Both A and B

26. What is the primary purpose of `groupby()` in Pandas?

Options:

- A) To sort data
- B) To filter rows
- C) To split data into groups and apply aggregation
- D) To merge datasets

Correct Answer: C) To split data into groups and apply aggregation

MATPLOTLIB AND SEABORN QUESTIONS

27. What will be the output of the following code?

```
import matplotlib.pyplot as plt
```

```
x = [1, 2, 3]
```

```
y = [4, 5, 6]
```

```
plt.plot(x, y)
```

```
plt.show()
```

Options:

- A) Scatter plot
- B) Line plot
- C) Bar plot
- D) Histogram

Correct Answer: B) Line plot

28.Which Seaborn function is used to visualize the distribution of a dataset?

Options:

- A) `sns.lineplot()`
- B) `sns.barplot()`
- C) `sns.histplot()`
- D) `sns.scatterplot()`

Correct Answer: C) `sns.histplot()`

29.What is the primary purpose of using a heatmap in data visualization?

Options:

- A) To display data using bars for comparison
- B) To represent data values as colors in a matrix for easy pattern recognition
- C) To show distribution of a single variable
- D) To plot relationships using points

Correct Answer: B) To represent data values as colors in a matrix for easy pattern recognition

30.Which type of visualization is shown below and what is its main purpose?

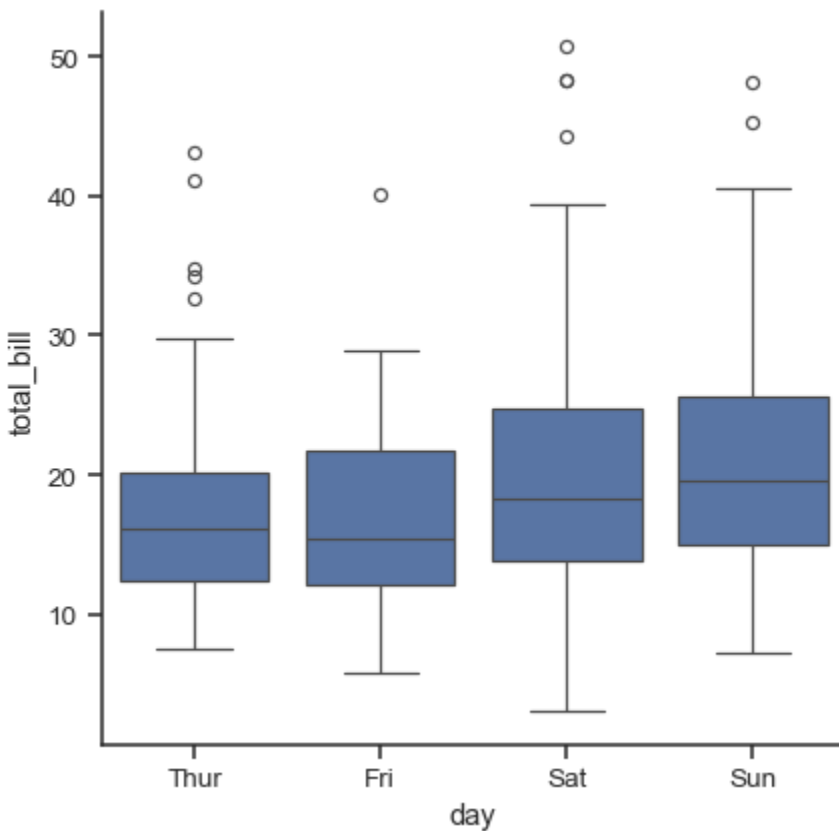


Options:

- A) Scatter plot – shows relationship between two variables
- B) Heatmap – shows values as colors in a matrix (often correlation)
- C) Histogram – shows distribution
- D) Line plot – shows trend over time

Correct Answer: B) Heatmap – shows values as colors in a matrix (often correlation)

31. Identify the type of graph shown below:



Options:

- A) Scatter plot
- B) Line plot
- C) Box plot
- D) Histogram

Correct Answer: C) Box plot

33. What is the purpose of using `sns.kdeplot()` in Seaborn?

Options:

- A) To display categorical data using bars
- B) To visualize the probability density (smooth distribution) of a continuous variable
- C) To show correlation between variables
- D) To create a scatter plot

Correct Answer: B) To visualize the probability density (smooth distribution) of a continuous variable

